

CS/SSW 555 Agile Methods

Cyber Guru Crew (Team 17)

Sprint 1

3/26/2024

Project Background Information

Selected Project Option Number: Option 3

Main Objective of Project:

Develop a mobile application that uses AI to detect lies through voice stress, facial expressions, and response timing analysis. This application is aimed at professional and personal use, as well as it promising to enhance truth verification processes with user-friendly features.

Team Members and Roles:

Name	Role
Octavio Morales	Software Developer
Brandon Lopez	Machine Learning Analyst
Bhagawat Chapagain	Software Developer
Edmund Yuen	Machine Learning Analyst
Yunfei Tang	UX/UI Designer

List & Description of Each Sprint:

Sprint #	Objective
Sprint 1	Project onboarding and setup: - Define database schema and set up the database - Choose and set up testing framework - Basic data collection for initial model training
Sprint 2	UI/UX development - Design user interface for the app - Implement user interactions and flow
Sprint 3	Machine Learning Model development: - Select appropriate machine learning model - Train the model with collected data - Evaluate the performance of the model
Sprint 4	Machine Learning Model development:

	• Fine-tune the machine learning model • Integrate the UI components with the machine learning model
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Project Links:

Type	URL
GitHub/GitLab/Etc	https://github.com/octavio-morales1/CS555Team17.git
Jira	https://stevens-team17.atlassian.net/jira/software/projects/KAN/boards/1#
Live Demo (if applicable)	N/A

Technology Stack Used:

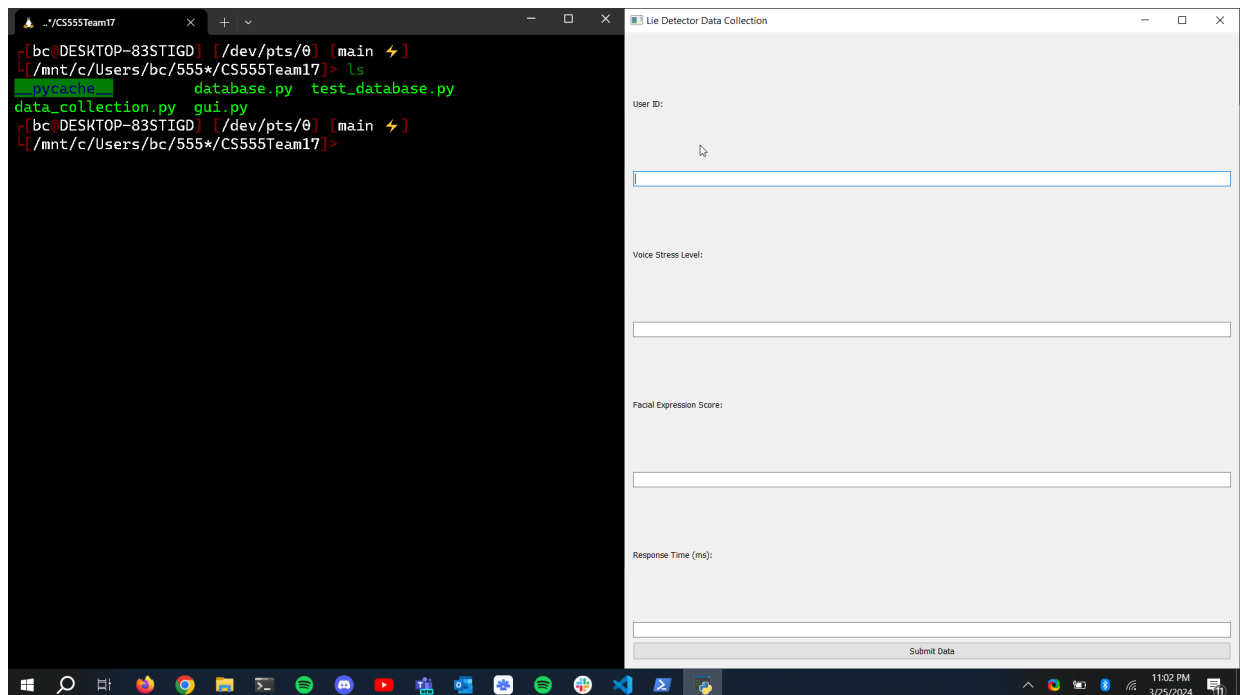
Type	Technology
Data Preprocessing	MatLab
UI/UX	PyQt5 (Python GUI)
Machine Learning	Kaggle
Database	MongoDB

Sprint Progress Update: Current Status

Summary of Work Completed

Below is a basic user interface for the product we are trying to build. It includes text boxes and corresponding labels for the information that is to be received. This will allow our target user base to use the application to detect lying through the use of these different values. Of course, the data itself must be harvested and inputted which will be done separately in another application. A feature we'd like to implement to encourage "flow" is the seamless transition between these two applications so that we are able to increase user productivity when using our application.

Pair programming was used to implement two different features. Functions called "initUI" and "onSubmit"; which allow the output of the user interface and then process the read data after all the boxes are filled by the end user. It was useful to be able to directly use fields defined in the first function in order to appropriately manipulate relevant data in the second function. The use of pair programming, in this case, was very successful as code that was being simultaneously written was highly compatible due to a causal relationship between code written in the User Interface function being used in the onSubmit function. This also falls into the Continuous Integration (CI), code that was written by different programmers was normally integrated into the same GitHub repository. The invocation of "ls" makes this clear with four different Python files "database.py", "test_database.py", "data_collection.py", "gui.py" were not all written by the same developer yet they are all within the same repository and some files are actually dependencies for code written by another.



Screenshot of Jira Backlog

Projects / CS SSW team / Sprint 3

Backlog

Epic ▾ Type ▾

Import work Insights View settings

▼ Sprint 1 20 Feb – 5 Mar (3 issues)

0 0 3 Start sprint

Project onboarding and setup: Define database schema and set up the database Choose and set up testing framework Basic data collection for initial model training

☑ KAN-6 Define database schema and set up the database Define database schema and set up the database	DONE ▾	=	
☑ KAN-7 Choose and set up testing framework	DONE ▾	=	
☑ KAN-8 Basic data collection for initial model training	DONE ▾	=	

+ Create issue

3 issues

▼ Sprint 2 6 Mar – 20 Mar (2 issues)

2 0 0 Start sprint

UI/UX development Design user interface for the app Implement user interactions and flow

☑ KAN-9 Design user interface for the app	TO DO ▾	=	
☑ KAN-10 Implement user interactions and flow	TO DO ▾	=	

+ Create issue

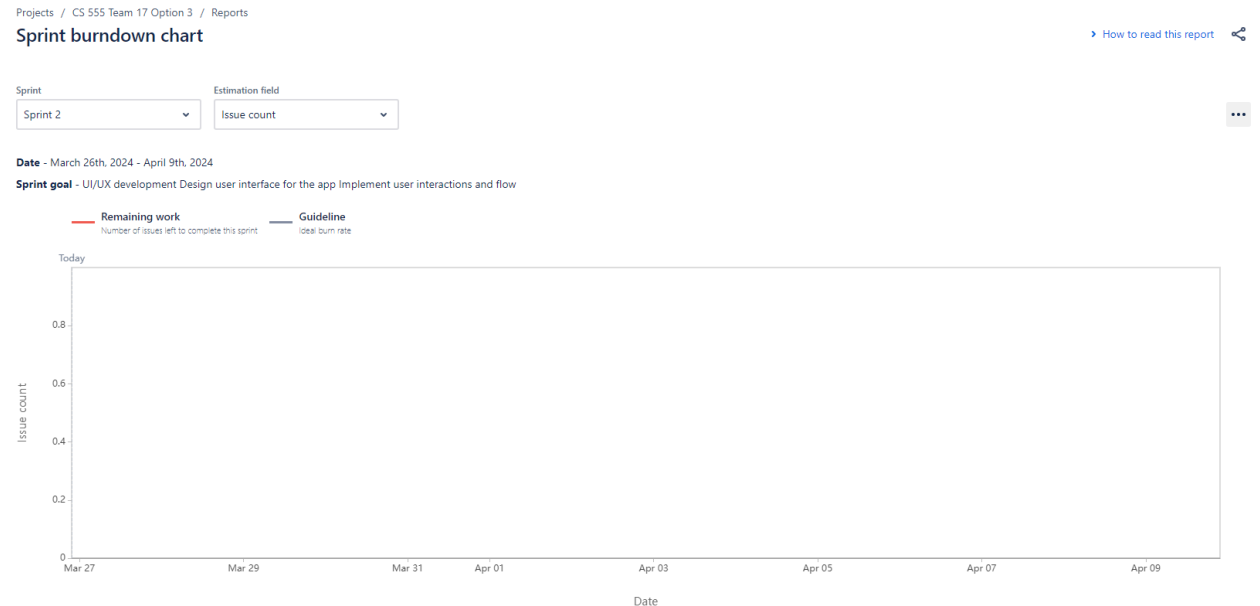
2 issues

▼ Sprint 3 21 Mar – 4 Apr (3 issues)

3 0 0 Start sprint

Machine Learning Model development: Select appropriate machine learning model Train the model with collected data Evaluate the performance of the model

Screenshot of Burndown Chart



As shown in the image, there are no indicators for the graph. We experienced some issues with the burndown chart. What it's supposed to look like is a gradual and constant decline in remaining work for the sprint, but in actuality there was no change until the last week, which is where you would see a gradual decline.

Video of Project at Current State:

It does not need to be one edited video. If there are multiple videos, please link all of them below:

Public Video URL (Google Drive, Loom, Etc)
No video present as this is not the current stage for visual/audio results.

Sprint Progress Update: Current Blockers

As of now, there are no blockers that are delaying the product of the project

Looking Ahead: Next Sprint

- Assess the selection process for the machine learning model to ensure the best fit for the project's needs. (KAN-11: Select appropriate machine learning model)
- Confirm the training phase has successfully used the collected data to create a robust model. (KAN- 12: Train the model with collected data)
- Evaluate the performance metrics of the trained model to determine its effectiveness and identify any potential improvements or adjustments needed. (Kan- 13: Evaluate the performance of the model)

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